TOPIC 2 Systems of Linear Equations and Inequalities

Name _____ Date _____

Reminders
Linear combinations: make one variable's coefficients opposites, then add the equations.
Multiply every term, including the number on the right side.
Fractions: multiply by the LCD first. Variable alone? Substitution is faster.

1. Solve: $x + y = 12$ and $x - y = 4$.

Tip: add the equations

A. (4, 8)

B. (16, −4)

C. (8, 4)

2. To eliminate y in $3x + y = 7$ and $2x + 2y = 6$, multiply the first equation by…

Tip: make the y-terms opposites

F. 2

G. 3

H. −2

3. Multiply $4x + y = 15$ by -2.

> **Tip:** multiply EVERY term

A. $-8x - 2y = -30$

B. $-8x + 2y = -30$

C. $-8x - 2y = 15$

4. Is $(6, -9)$ a solution of $4x + y = 15$?

> **Tip:** substitute both values

F. No, because $33 \neq 15$

G. Yes, because $15 = 15$

H. No, because $6 \neq -9$